

Self - Attention Matrix Summary

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- Input - 3 words, $x = (3, 512)$
- Attention learns three matrices w_q, w_k, w_v so w_q have $512D \rightarrow 64D$ $(512, 64)$
- $q = x * w_q = (3, 512) * (512, 64) = (3, 64)$
- Attention Score = $q k^T = (3, 64) * (64, 3)$
 $= (3, 3)$
- Scale - Score = Scores / $\sqrt{d_k}$ = Scaled values
- $\text{Softmax}(\text{Scores} \rightarrow \text{Probability}) = \text{Softmax}(\text{Scaled-Score})$
- Attention output = $\text{Softmax}(q \cdot k) * v$
 $= (3, 3) * (3, 64) = (3, 64)$

$$\# \text{ Attention}(q, k, v) = \text{Softmax}\left(\frac{q k^T}{\sqrt{d_k}}\right) * v$$

- In multihead attention we get 8 attention head so the output we get is $(3, 8, 64)$ or $(3, 512)$. In multihead attention we also do linear projection for combining into from all 8 heads $\rightarrow (3, 512) * (512, 512) = (3, 512)$